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New biosensors

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23.1 Introduction

Biosensors are one of the foremost applications of biotechnology of relevance to the food industry. These devices have the ability to provide rapid, cost effective, specific and reliable quantitative analysis. However, the variety of biological complexity of molecular structures and the wide range of concentrations coupled with a complex matrix are some of the bioanalytical challenges for food analysis. Recently farmers and food manufacturers have been subjected to enormous pressure to comply with legislation and satisfy consumers. Increased demands for organic food, cook-chill food and fresh food, free from traces of chemicals and pathogenic organism have intensified the need for better diagnostics methods for real-time analysis. Immunoassays and polymerase chain reaction (PCR) techniques are now available for a range of food analysis, but more rapid and easy methods are required such as biosensors and affinity sensors. A range of biosensor instruments is required for the agro-food market and these include multi-analysers, bench-top portable instruments to one-shot disposable sensors.

A wide variety of biosensor devices exploiting a range of biological and synthetic receptors in combination with a range of transducers are reported in the literature (Turner, 1999). Within each combination lies a range of alternative strategies and approaches that can be applied to the food industry and process engineering. However, the commercial success of biosensors is limited to a small number of applications and where the market size justified significant investment in research and development. Success has been achieved with individual sensors, which are supported by other systems such as electronics, fluidics and separation technology. To obtain functional sensing devices with the required performance and cost, several other components and technologies are needed. Membranes can be used in biosensors as a retention barrier for the biological component whilst allowing the analyte to pass, exclusion of interferents on the bases of size or charge, or acting as a mass transport barrier to extend the linear range of the biosensor. Receptor immobilisation on the sensor surface is a crucial step in achieving a successful device.

The successful development of commercially viable biosensors can be realised using advanced fabrication techniques. Screen-printing (a thick-film process) has been used to

Table 23.1 Potential market for biosensors

Application	Market ($ billion)	Source
Microsystems market for *in-vitro* diagnostics	19	Marshall (1999)
Drug testing	1.9	Marshall (1999)
Drug delivery	1	Marshall (1999)
Military (warfare agent detection)	0.64	Anonymous (1998a)
Electronic noses and tongues (health care, testing of product quality, to establishing authenticity of perfumes and wines)	4	Anonymous (1998b)

mass produce large numbers of reproducible, inexpensive devices rapidly. The commercial success of many biosensors such as the MediSense (now Abbott) pocket-sized glucose-sensing devices relies on such methodology. Ink-jet printing is suitable for the deposition of droplets (less than 1nl) with high speed. Automated syringe-type techniques (such as Cavro deposition) can be used to fabricate biosensor devices. Lithographic techniques can produce small structures with well-defined geometries. The biointerface between the analyte/receptor and transducer is also a key element in designing a successful biosensor. The interface can be made more selective thus reducing interference compounds.

The food technologist has an interest in generally increased sensitivity and limits of detection for a range of analytes. However, the reliable detection of trace quantities of additives, contaminants and other chemicals is the first priority. The potentially huge biosensors market (Table 23.1) can be attributed to these problems. At present less than 10% of this market has been exploited with two-thirds of sales accounted for by glucose and BOD sensors (Crabb, 1998). Companies such as Abbott, Boehringer Mannheim, Bayer and LifeScan now have competing mediated biosensors and are rapidly displacing conventional reflectance photometry. Biosensors specific for food testing are now available on the market. Immunosensors and other affinity sensors offer an ultra-low detection limit and electrochemical detection may be readily integrated with chromatographic techniques to yield user-friendly devices. The diagnostics field is undergoing rapid changes and investment is rocketing worldwide.

23.2 Novel sensing receptors

Enzymes are the most commonly used receptors in biosensors. However, a range of biological molecules and biomimics have also been developed and used as the sensing receptors (Toko, 2000; Turner, 2000). Advances in antibodies and recombinant antibody production has created a new range of immunosensors devices to replace immunoassays and dipstick assays. Antibody fragments and molecularly engineered antibodies with an enhanced affinity and stability is a growing area of research for new receptors. Biological components (i.e. antibodies, cell receptors, DNA, micro-organism) offer an exquisite sensitivity coupled in most cases with high selectivity but suffer from poor stability. The use of chemical stabilisation and immobilisation procedures such as sol gels as an immobilisation matrix has been demonstrated recently as a good stabilisation strategy (Newman *et al.*, 1998; Tothill and Turner, 1998). However, some biosensor applications require stability far beyond what can be exhibited by biological molecules. Particular

attention has been paid to solving the most critical problems related to the application of biological molecules in sensing:

- low stability
- high cost of enzymes and receptors
- poor performance in organic solvents, at low and high pHs, and at high temperature
- absence of enzymes or receptors able to recognise certain target analytes.

Research into the use of biomimics is increasing. The artificial receptor approach is designed to overcome these difficulties. The search for possible solutions to the low stability of biological molecules has taken several directions:

- use of molecules produced by extremophiles, e.g. microorganisms growing in hot springs
- improvement of biomolecule stability using genetic manipulation
- substitution of enzymes, receptors and antibodies with artificial analogues synthesised using combinatorial chemistry or rational design
- synthesis of synthetic receptors using molecular imprinting.

The biotechnological approach involving the use of molecular biology to modify natural receptors with enhanced properties is a more classic approach. However, success in this area has been scarce. New advances in receptor discovery such as the development of artificial receptors using the combined approach of molecular modelling and molecularly imprinted polymer (MIP) or combinatorial synthesis promise to revolutionise the range of receptors suitable for biosensor construction.

23.2.1 Combinatorial chemistry

Combinatorial chemistry is used to synthesise libraries of compounds for the screening of novel affinity ligands that can be used for sensor applications. This is a rapidly developing field of receptor discovery. The use of peptide libraries with natural and unnatural amino acids as building blocks is the present approach. Various types of resins such as Wang, Trityl and PEGA and Merrifield are used to prepare the libraries. The libraries are then screened and positive results visualised using a microscope (Fig. 23.1). The use of the molecular modelling approach to rationally design affinity ligands and guide the selection of the building blocks to be used in combinatorial synthesis is one of the most recent advances in receptor discovery. Molecular modelling will also visualise and evaluate the interactions between the target analyte and the potential receptor. Molecular modelling is applied to provide structural information for the target analyte that can direct the design of combinatorial libraries and rational design of the artificial receptor (Chen *et al.*, 2001).

Combinatorial synthesis is a widely used method for receptor production for the drug discovery arena. However, its use for the production of receptors for biosensors is still in its early stages. The technology is expensive and needs investment in equipment and personnel.

23.2.2 Molecular imprinting

The molecular imprinting approach involves formation of a complex between functional monomers and the guest molecule (template) in an appropriate solution, and ‘freezing‘ of this complex by polymerisation in the presence of a cross-linker. Following removal of

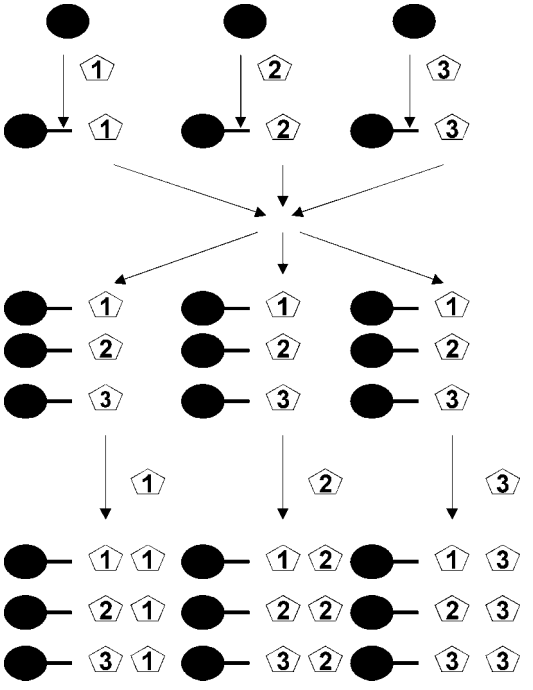


Fig. 23.1 Schematic of split-and-mix combinatorial synthesis.

the template by washing, binding sites (imprints) are left within the polymer with a structure, which is complementary to the template molecule (Fig. 23.2). The theory of imprinting polymerisation and practical aspects of MIP application are covered in several recent reviews (Wulff, 1995, Mosbach *et al*., 1998, Piletsky *et al*., 1999a). The essential advantages of molecularly imprinted polymers (MIPs) are:

- MIPs can be prepared for a broad variety of analytes
- MIPs have good affinity and specificity, comparable to natural receptors
- MIPs are stable in organic solvents, at low and high pH, and at high temperatures
- Polymers should be easily integrated into the microfabrication process.

Imprinted polymers can in theory be prepared for any kind of substances. Examples of successful imprinting include inorganic ions, drugs, nucleic acids, proteins and even cells. The high specificity and stability of MIPs render them promising alternatives to the enzymes, antibodies, and natural receptors used in sensor technology (Jenkins *et al.*, 1997). The main limitations associated with development of MIP sensors are: (a) absence of a general procedure for MIP preparation and (b) difficulty in transformation of the binding event into an electric signal.

Several attempts have been made to develop a rational procedure to design MIPs with a certain level of specificity (Takeuchi *et al*., 1999). Recently we proposed a method that we believe to be a general solution for MIP design. It includes computational screening of a virtual library of functional monomers against a target molecule followed by the selection of those with the highest binding score. The software facilitates calculations using different dielectric constants to reflect the polarity of the environment (solvent) in which the polymers have to be prepared or used.

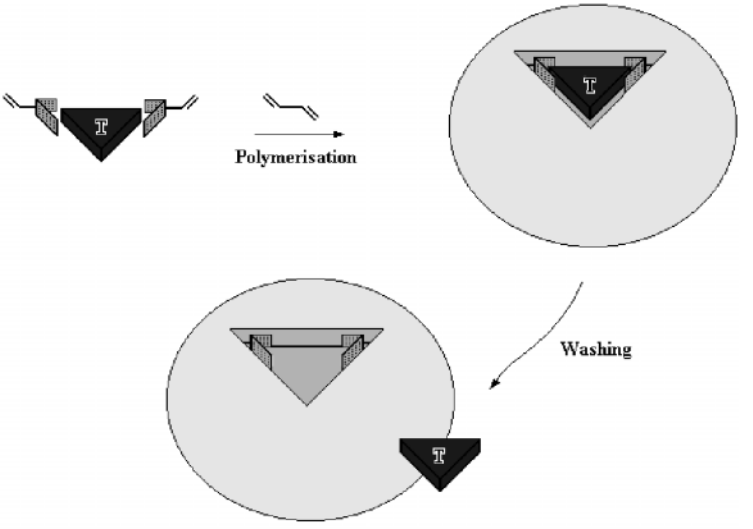


Fig. 23.2 Scheme illustrating molecular imprinting polymerisation.

Design of MIP sensors

Two general principles have been used for the design of MIP sensors: (a) substitution of antibodies in immunosensors by MIPs (affinity sensors) and (b) exploitation of the receptor properties of the imprinted polymers (receptor sensors). Depending on the properties of the template, detection can be achieved with the help of conductometric, amperometric or optical devices integrated with MIPs using photodeposition (Kroger *et al*., 1999), grafting (Mirsky *et al*., 1999) or electropolymerisation (Panasyuk *et al*., 1999).

MIP-based affinity sensors can be used to detect a template, which possess specific properties such as optical absorbance, fluorescence, or electrochemical activity. A typical example is monitoring fluorescent dansyl-L-phenylalanine adsorbed by the imprinted polymer and detected using a fibre-optic sensing device. In another example, fluorescent polycyclic aromatic hydrocarbons were selectively enriched and detected using optical sensors based on imprinted polyurethanes. The direct detection of 'inert' templates can be realised in receptor sensors by using the MIP's ability to change conformation upon binding with a template, leading to a change in the measuring property (conductivity, permeability or surface potential). Sensors, specific for L-phenylalanine, cholesterol, sialic acid and atrazine (Piletsky *et al*., 1998, Sergeyeva *et al*., 1999) show high selectivity and sensitivity in the micromolar-nanomolar range.

Dickert and Thierer (1996) proposed an interesting approach to the design of signalling MIPs and their use in sensors for organic solvents. They used the environmentally sensitive functional monomer, 3,3-diphenylphtalide, integrated into a cross-linked matrix and capable of changing its optical properties in the presence of a template. A similar approach has been used in the design of a fluorescent sensor for cAMP detection. In this case a fluorescent dye, trans-4-[p-(N,N-dimethylamino)styryl]-N-vinylbenzyl pyridinium chloride, was co-polymerised with a cross-linker and template to display two functions simultaneously; template recognition and sensing. Murray and co-authors developed very sensitive lanthanide-based luminescent sensors for sarin and soman with a detection limit of 125 ppt (Jenkins *et al*., 1997).

Despite the fact that essential progress has been made in the field of MIP catalysts

practically no examples of MIP catalytic sensors are available. It is anticipated that with further success in preparation of effective MIP-based catalysts, reacting with water-soluble and practically important analytes, this situation will change. The list of MIP sensor prototypes developed over the last two decades is quite extensive (Table 23.2). In many cases the sensitivity and selectivity levels achieved were high enough to justify their practical application.

Future perspectives for MIP

MIP sensors have their own niche for potential applications:

- control of industrial processes with extreme pH, pressure, temperature, use of organic solvents
- control of biotechnological processes where sensors should be autoclaved
- continuous on-line monitoring
- space and military applications
- robust multisensors (biochips).

Promising application areas for MIP sensors are food, medicine and the pharmaceutical industry where the stability of MIPs, e.g. during sterilisation, would be of significant advantage. The industrial use of MIP sensors under extreme conditions is also particularly attractive. Although this is not a big market, it is important due to the absence of competition from traditional biosensors. Both industry and environmental agencies have a need for remote and stable MIP sensors for continuous monitoring of possible emissions and chemical spillage.

The preliminary stage of the evaluation of MIPs as promising materials suitable for sensor technology is complete. Several small and medium-size companies in the US

Table 23.2 Application of MIPs in sensor technology

Template	Detection	Detection limit, ppm	Reference
Amino acids and their derivatives	Electrochemical	~500	Panasyuk *et al*., 1999
	Optical	5	Piletsky *et al*., 1999b
		30	Kriz *et al*.,1995
Drugs	Electrochemical	3.5	Mirsky *et al*., 1999
	Optical	~1	Lai *et al*., 1998,
		3	Levi *et al*., 1997
Gases	Electrochemical	5	Hernandez and Bachas, 1998
Herbicides	Electrochemical	~1	Kroger *et al*., 1999
		0.005	Sergeyeva *et al*., 1999
Inorganic anions and cations	Electrochemical	50	Hutchins and Bachas, 1995
		10	Murray *et al*., 1997
Nucleic acids and derivatives	Electrochemical	~12	Piletsky *et al*., 1990
	Optical	1	Turkewitsch *et al*.,1998
PAHs	Optical	<0.001	Dickert *et al*., 1998a
Phenols	Electrochemical	~1	Morita *et al*., 1997
Sarin, Soman	Optical	$7\ 10^{-6}$	Jenkins *et al*., 1997
Solvent vapours	Optical	2000	Dickert and Thierer, 1996
	QMB	0.1–4	Dickert *et al*., 1998b
Sugars and their derivatives	Electrochemical	~1	Piletsky *et al*., 1998
	Optical	~1	Piletsky *et al*., 1996

Note: QMB, quartz microbalance

and Europe are now working actively to commercialise the first generation of MIP sensors which will replace their more traditional enzyme and antibody-based predecessors.

23.3 Sensor arrays

The formulation of arrays of sensors to diagnose one sample for several analytes is a very attractive approach. These types of sensors are available as bench-top instruments for critical care and food analysis and mainly feature biosensors for glucose, lactate, urea and creatinine. The demands for detection of a range of analytes are increasing.

Miniaturisation of biosensors is also being driven to micro-array systems. The use of photolithography, self-assembly and microcontact printing offer routes to high-density arrays. The driving force behind this comes from the high throughput-screening programme within the drug discovery businesses. The application of silicon fabrication technology for biosensor array production is receiving great attention from diagnostics companies. The attractive issue here is the on-chip micro-fabrication and electronic signal amplification and processing. A disposable DNA probe array (GeneChip) has been developed by Affymetrix (USA). These DNA chips are used for microorganisms detection and identification.

A wide variety of computational intelligence techniques are being used in conjunction with sensors and biosensors. These include neural networks (ANNs), mathematical modelling, chemometrics and data mining. Data analysis and bioinformatics are very important to make the most of the data collected from sensor responses. Sensing possibilities based on fingerprinting type approaches such as the Noses and Tongues array sensing are being developed rapidly.

23.4 The electronic nose instrument

Generic electronic noses (e.nose) or odour mapping systems have been defined as ‘an odour, which is presented to an active material of a sensor which converts a chemical input into an electrical signal’. The construction of sensor arrays with partial sensitivity to different groups of volatiles means that a broad range of applications can be considered both in the environment and for food. There are ranges of types of sensor arrays, which are now being used in e.nose systems. These include inorganic semi-conducting materials such as oxides and catalytic metals, which are used at elevated temperatures and are good for alcohols, and conducting polymers which are polypyrrole, polyaniline-based which can operate at room temperature with a sensitivity of 0.1–100 ppm of volatiles on piezoelectric and electrochemical sensors. A range of commercial products is now available on the market with food, environmental and medical applications. Some of the available e.nose systems available are listed in Tables 23.3 and 19.2.

Knowledge of the potential use of electronic noses or odour mappers for food applications has increased rapidly during the last decade. Some recent examples of e.nose applications in the food industry include direct monitoring of off-odours of food products, or differentiation of different varieties or types of related product, e.g. coffee, olive oil, cheeses, fruit ripeness, to early detection of the activity of spoilage microorganisms. Table 23.4 lists some of the areas where applications of e.nose technology have been considered or made. There are a number of common simple chemical odours, which are

Table 23.3 List of some commercial electronic nose systems available today (see also Table 19.2)

Electronic nose	Company	Description
Portable odour monitor	Cyrano Sciences Ltd, USA	Hand-held device with up to 32 sensors; surface polymer carbon sensors
E.Nose 3000	Marconi Instruments Ltd, UK	Metal oxide sensors; 8–32 sensors
Aromascan	Osmetec Ltd, UK	32 surface polymer sensor array
BH114	Bloodhound Sensors Ltd, UK	14 surface polymer sensor array
Fox 3000	Alpha Mos, France	Large laboratory systems with 12 sensors
NST 3320	Nordic Sensor Technologies, Sweden	Metal oxide sensor based, automated system
MGD-1	Environics Ltd, Finland	Based on patented IMCELL^{TM} measuring method with 6 channels, for gas sensing

indicative of changes in quality in a range of food products. Table 23.5 lists some examples of important odours, and their presence in different products. This is indicative of the fact that qualitative changes in patterns of volatiles can be used and an indicator of changes in quality and perhaps in quality assurance. Of particular importance are not just the measurement but the analyses of the data. A wide range of statistical procedures, and neural network training has to be carried out to enable an e.nose system to be used effectively, especially for industrial applications.

Table 23.4 Recent applications of electronic nose technology in the food industry (see also Chapter 19)

Applications	References
Classification of food spoilage bacteria	Pavlou *et al*. (2000); Magan *et al*. (2000)
Monitoring of packaging quality	Holmberg *et al*. (1995); Forsgen *et al*. (1999); Culter (1999)
Monitoring of agricultural malodours	Misselbrook *et al*. (1997)
Fish freshness	Di Natale *et al*. (1997)
Meat freshness	Eklov *et al*. (1998); Blixt and Borch (19990
Classification of vegetable oils	Martin *et al*. (1999)
Italian vinegar classification	Anklam *et al*. (1998)
Cheese maturity	Schaller *et al*. (1999)
Classification of fruit and fruit ripeness	Llobet *et al*. (1999); Hines *et al*. (1999); Sinesio *et al*. (2000)
Detection of off-odours in sugar	Kaipainen *et al*. (1997)
Classification of beer and tainting	Pearce *et al*. (1993); Tomlinson *et al*. (1995)
Authentication of wine	Di Natale *et al*. (1996); Di Natale and D'Amico (1998).
Differentiation of coffees	Gardener *et al*. (1992)
Mycological quality of cereal grains and bread	Borjesson *et al*. (1996); Keshri *et al*. (1998); Schnurer *et al*. (1999); Magan and Evans (2000); Keshri and Magan (2000).

Table 23.5 A list of key odours that can be used as indicators of changes in food products

Chemical type	Odour type and product
Diacetyl	off-odours of beer
Trans-2-hexenal	green leaves
Geraniol	rose
5-isopropyl-2-methylphenol	thyme
Limonene	lemons
Cis-4-heptanal	off-flavours of white fish
Octa-1,5-diene-3-1	off-flavours of butter
2-Isobutyl-3-methoxypyrazine	green peppers
alpha-Terpinethiol	grapefruit
1-octen-3-ol	fungal spoilage

23.5 Genetically modified food

Genetically modified organisms (GMOs) have a crucial feature of an altered genome leading to production of unique protein(s). Detection of these organisms is therefore focused on either detecting the altered gene(s) or the novel protein(s) (Brett *et al*., 1999). In Europe the introduction of this type of food falls under the Novel Food Regulation. An essential component of testing for novel foods is the availability of reliable analytical methods with proven performance. The first transgenic food crops like soybeans and Maize are on the market and others such as tomatoes, squash, potatoes, and also cotton are now appearing on the market. Other transgenic crops such as sugar beets, sunflower, rice and wheat are being developed. Gene technology has been applied to obtain improved functional properties, such as starch potatoes with amylopectin and without amylose and tomatoes without pectolytic enzymes. Also crops with improved yield or pest resistance as in the case with Roundup Ready (RR) soy beans (Monsanto) and the Bt resistant maize (Novartis) are on the market (Duijn *et al*., 1999). To date consumers have stated that they want clear and concise information on the GMO content in food. Food processors need meaningful and unambiguous testing results in order to comply with labelling requirements.

23.5.1 DNA-based methods

Surveillance of food labelling with respect to genetically modified organisms (GMOs) may require DNA-based diagnostics methods. The polymerase chain reaction (PCR) is a highly specific and sensitive method for the detection of DNA or RNA. A vast number of PCR systems for food control exists, such as for the detection of pathogenic microorganisms of food components as well as for the identification of animal species in meat products (Meyer and Jaccaud, 1997; Hübner *et al*., 1999). PCR also provides a reliable tool for the identification of GMO-containing food in order to control food labelling regulations concerning GMOs. Some of the detection methods for GMO-containing food is based on a qualitative PCR system specific for the 35S-promoter originating from the Cauliflower mosaic virus (35S-PCR), NOS terminator or the Agro-bacterium CP4 synthase gene. Although current GMO detection methods are feasible for food control, inter-laboratory reproducibility remains unsatisfactory. Quantitative competitive PCR (QC-PCR) has been found to give reproducible results for GMO

content of food samples (Hübner *et al*., 1999). Development of more advanced multi-detection systems is needed.

DNA biosensors are currently used in medical diagnostics but their application in food analysis is expanding. The Affymetrix GeneChip, which based on DNA probe array, is used for microorganisms identification or detection of GMO food. GeneScan Europe (Germany) markets test kits for GM food detection in human and animal foods based on DNA analysis using PCR.

23.5.2 Protein-based methods

Detection of the novel proteins in GMOs is usually based on immunoassays (Stave, 1999) or the use of antibodies combined with Western blotting analysis (Duijn *et al*., 1999). It is very important to understand how the protein behaves in food production and processing. The protein may undergo thermal and enzymatic degradation, giving rise to peptide fragments derived from the original GMO protein. Also the homology of the novel protein sequence with other proteins in the food sample may hinder analysis. Choosing a specific peptide sequence from the novel protein that is more specific for this target protein may help in overcoming these difficulties. Changes in the conformation of proteins during food processing can lead to reduction in the antibody recognition of the target protein. Antibodies directed towards processing stable epitopes of the protein sequence will then be required to reduce these problems. The detection of modified Soya protein in food is usually carried out using immunoassays (McNeal, 1988, Brett *et al*., 1999). Developing immunosensors should not be difficult from available immunoassays for GMOs. This will indicate progress for GMO protein analysis. One point that needs to be emphasised is that most novel proteins introduced into transgenic crops are proprietary to the company developing GMO food. These companies need to be obliged to make antigens and antibodies, or the means to produce them, available to outside parties for test development.

23.6 Commercial biosensors

The biosensor market is still dominated by the glucose biocatalytic sensor for medical application. A small number of companies are in control of this market and these include Abbot, Bayer, Roche Diagnostics (Boehringer) and LifeScan (Inverness Medical) (Newman *et al.*, 2000). Other significant markets are the environment and food, which remain the smallest due to low investment. To date a range of biosensors have been developed for food analysis (Scott, 1998) but most are still in the research stage. Table 23.6 lists some of the commercially available biosensors for food analysis. The YSI 2700 SELECT^{TM} is used for on-line fermentation monitoring or off-line sample analysis for food and drinks. The biosensor system can provide multi-analytes analysis for the food industry.

Immunoassays for the detection of substances in food are well established. Applications include detection of hormones, toxins, allergens, pesticides, micro-organisms, antibiotics, and authenticity and adulteration of food with other food substances. A number of books have been published regarding the use of immunoassays in foods (Beier and Stanker, 1996; Morgan *et al*., 1992). This indicates the wide range of antibodies available for food analysis, which can be implemented in immunosensors format (Rogers and Mulchandani, 1998). A range of biosensors and immunosensors are

Table 23.6 Biosensors for food and agricultural analysis

Company	Sensor	Description	Application
ABTECH Scientific Inc. (USA)	ToxSen™ EnVOCSys™	Chemical toxicity sensor for environmental monitoring. An environmental sensor array for gas phase analysis.	On-site screening of chemical toxicity of water, wastewater, and effluent streams. Volatile organic compounds emissions from industrial sources.
Affymetrix Inc. (USA)	Gene Chip probe array	DNA probe for food and environmental monitoring.	Microorganism contamination of food and environmental samples.
Biacore AB (Sweden)	BIACORE 3000	Surface plasmon resonance affinity based biosensor can be used for food and environmental analysis.	Microorganisms, drugs, carbohydrates vitamins and other low molecular weight molecules.
TRACE Biotech AG Germany)	AlcoTrace GlucoTrace	Enzyme biosensor for food and environmental analysis. Enzyme biosensor	Soft drinks and non-alcoholic beer, wine and champagne. Glucose analysis in nutrient media and biotechnological processes.
Yellow Springs Instruments (USA)	YSI 2700 SELECT™	Enzyme based biosensors for food analysis.	Glucose, lactate, lactose, sucrose, galactose, glutamate, ethanol, glutamine, methanol, starch.
Molecular Devices Corporation (USA)	Threshold® workstation	Affinity capture and light addressable potentiometric sensor (LAPS).	Micoorganisms detection based on DNA assay. Immunoligand assay for analyte detection.
Sensor Solutions Ltd (UK)	Beer Alcohol Analyser	Enzyme electrodes with novel membranes.	Food and drinks market (under development).
Texas Instruments (USA)	TI sensor system	Surface plasmon resonance affinity based sensor.	Affinity interactions and DNA analysis.

being developed today to sense these compounds in food, soil and water samples and heavy metal ions in food samples. All of these tests can be modified into immunosensors and affinity sensors. The BIA core™ biosensor system based on SPR technology is also being used for food analysis such as the detection of microorganisms and vitamins in food samples.

Analytical methods for authenticity of food testing have been described for all types of foods. Highly discriminating are proteins and nucleic acids. The identification of meat species using protein-based analytical methods is much more easy to perform and also sensitive. The use of antibodies directed against heat-denatured meat proteins is an interesting development (Lumley, 1997). DNA-based methods for meat species identification has been developed in recent years.

23.7 Conclusion

Biosensor technology is a rapidly expanding area, as evidenced by the quantity of patents published, which has doubled during the last five years. The majority of these patents describe innovations in materials science rather than new types of transducers or detection principles. Particular attention has been paid to solving the most critical problems related to the application of biological molecules in sensing. New biosensors are appearing on the market as a result of the large investments being directed at niche areas.

Analysis of food can be a complex issue due to the diversity of food matrices. This places an additional strain on analysts to come up with the required sensor performance. Test developers must have incentives to invest in future technology development and testing infrastructure for food analysis. Regulators need to implement compliance monitoring in accordance with the EU's international trade obligations. Lastly, food is an essential commodity for the health and sustenance of the human race and looking after the food we consume must be everyone's main priority.

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